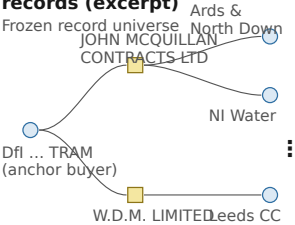


Example: a three-hop procurement query

Follow DfI Transport and Road Asset Management TRAM's lead suppliers outward: setting DfI Transport and Road Asset Management TRAM itself aside, which buyers do those same suppliers also work for?

Observed buyer-supplier records (excerpt)



Record projection

buyer	supplier	contract suffix
DfI ... TRAM	JOHN MCQ.	...030700
Ards & N. Down	JOHN MCQ.	...011224
NI Water	JOHN MCQ.	...004375
DfI ... TRAM	W.D.M.	...013310
Leeds CC	W.D.M.	...017146

○ buyer value ■ supplier value

(a) Direct plan generation (base model)

$VALUES_contract_id[supplier_name = k] \rightarrow VALUES_contract_id[buyer_name = k] \rightarrow []$

buyer/supplier role reversed

$k = \text{DfI Transport and Road Asset Management TRAM}$ wrong return field

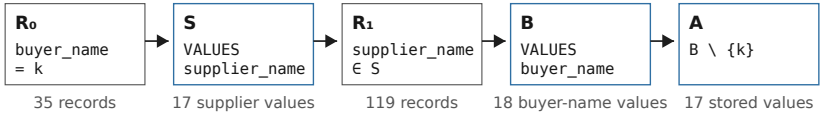
✗

(b) Typed compositional query

$S = VALUES_supplier(FILTER_buyer=k(R))$

$validate_tree(A) \rightarrow VALUES \checkmark$

$A = VALUES_buyer(FILTER_supplier \in S(R)) \setminus \{k\}$



17 stored buyer-name values

Ards and North Down Borough Council
Leeds City Council · NI Water and its subsidiaries · Education Authority NI · ...

exact oracle-set match ✓